Introduction to GPU Programming

cuSPARSE Syntax

cuSPARSE High Level Description

- A library that manages sparse matrices
- Helper, Level-1 to Level-3, and conversion functions
- Level 1 functions are all deprecated, don’t use them as they will be removed in the next major version of CUDA
- Level 2 functions perform operations between sparse matrices and dense vectors
- Level 3 functions perform operations between sparse matrices and dense matrices

[1]

Level-2 Operations

- cusparse<t>bsrmv() – BSR storage based multiplications
- cusparse<t>bsrsv2_bufferSize - buffer size for bsrv2 linear system
- cusparse<t>bsrsv2_analysis - bsrv2 analysis phase
- cusparse<t>bsrsv2_solve - bsrv2 solve phase

Level-3 Operations

- cusparse<t>bsrmm() – BSR storage based matrix – matrix multiplication
- cusparse<t> bsrmm_bufferSize - buffer size for BSR
- cusparse<t> bsrmm _analysis – bsr matrix – matrix analysis phase
- cusparse<t>bsrmm_solve – bsr matrix – matrix solve phase

References

- [1] 2021. NVIDIA. CUDA Developer Toolkit Documentation – cuSPARSE Library. Found at https://docs.nvidia.com/cuda/cusparse/index.html.

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